



Data Structures
CS 246 - 001
Department of Physics and Computer Science
Medgar Evers College
Exam 1 - MakeUp

Instructions:

- Each task requires creating a header file that defines classes, functions, and objects based on a case study.
- The header files must include a header guard, define classes and objects within the namespace *dse*.
- Only libraries 'iostream', 'string', 'iomanip', 'sstream', 'stdexcept', 'Object.h', 'Array.h', and 'Comparable.h' can be included in any header file.
- Documentation for 'Object.h', 'Array.h', and 'Comparable.h' are provided.
- All classes created must define essential methods publicly and fields privately.
- All classes, excluding interfaces, must define their special member functions, which must be public.
- All interfaces must define their destructor.
- All derived classes must have a virtual destructor.
- Notes are not allowed.
- Cheating of any kind is prohibited and will not be tolerated.
- **Violating and failing to follow rules will result in an automatic zero (0) for the exam.**

TO ACKNOWLEDGE THAT YOU HAVE READ AND UNDERSTOOD THE INSTRUCTIONS ABOVE,
PRINT YOUR NAME AND THE DATE ON BOTH THIS SHEET AND THE BLUE BOOK

Name: _____

Date: _____

Grading

Problem	Maximum Points	Points Earned
01	4.0	
02	4.0	
Total	8.0	

1. Case Study Name: Pairing

There are several objects with two aspects. Such objects have the characteristics:

- The types of its aspects can be different.
- The content of its aspects can be modified and viewed.

Using the specifications above, implement a generic class with two generic types *Pair* that provides:

- A default constructor that initializes its aspects to their default values.
- A getter method for each aspect.
- A virtual setter method for each aspect.

2. Case Study Name: Web Credentials

Credentials for a website require a username and a password, such that:

- Usernames are alphanumeric and at least 5 characters long.
- Passwords are at least 8 characters long with at least 1 letter and 1 digit.

Using the specifications above, create a class named *Credential* that public inherits string-string *Pair* and contains:

- A default constructor that assigns "admin" and "ABC12345" to the username and password aspects.
- A overridden setter method for the username that only performs the assignment if the parameter adheres to the username constraints.
- A overridden setter method for the password that only performs the assignment if the parameter adheres to the password constraints.